Eataly: A Decentralized Campus Food Management Ecosystem with Load-Aware Routing, Capacity-Bounded Scheduling, and On-Device Demand Forecasting

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***Abstract*—Collegiate dining ecosystems face severe structural queuing bottlenecks caused by timetable-synchronized demand, where thousands of students and faculty members converge upon limited kitchen kiosks during short inter-lecture intervals. Conventional mobile ordering applications fail to alleviate this friction; instead, they merely displace physical lines into unmanageable kitchen backlogs. This paper presents *Eataly*, a decentralized campus food management ecosystem structured around five core research contributions: (i) an optimization-driven *Smart Kitchen Load Balancer* that jointly minimizes queue drain latency, walking time, and forecast kitchen strain to steer foot traffic; (ii) a *Capacity-Bounded Slot Scheduler* featuring an automated three-tier conflict resolution engine (slot temporal shifting, cross-canteen rerouting, and transactional waitlisting) that preserves promise credibility without rejecting orders; (iii) the *Adaptive Learning and Intelligent Demand Prediction Engine (ALIDPE)*, an on-device seasonal smoothing module executing on edge tablets to forecast break-level item demand and optimize perishable inventory; (iv) an offline-first transactional pipeline using optimistic synchronization; and (v) an operational accessibility architecture comprising a bilingual (English/Tamil) kitchen management console and decentralized campus pickup hubs. Controlled simulations across three collegiate dining facilities (180 orders per peak rush) demonstrate that Eataly reduces counter queues from 5.51 min down to 0.04 min, achieves near-optimal load distribution (Jain's index 0.99), and slashes demand forecasting error (WAPE) to 17.7% relative to rolling moving-average baselines.**

***Index Terms*—Campus logistics, queueing theory, load balancing, capacity-bounded scheduling, edge demand forecasting, offline-first mobile architecture, assistive user interfaces.**

I. Introduction

Collegiate dining facilities represent an acute instance of synchronized temporal congestion. Unlike urban commercial eateries that handle continuous stochastic arrivals throughout the day, academic institutions operate under rigid bell schedules. The simultaneous release of thousands of students and staff during fixed 15- to 45-minute inter-lecture breaks induces extreme arrival spikes across a finite set of preparation kiosks [1], [2]. Because culinary facilities cannot scale physical preparation counters or kitchen personnel dynamically for a transient 20-minute rush, severe operational collapse inevitably ensues: students endure intolerable wait times that consume their designated rest periods, kiosks face unexpected stock-outs, and unguided production yields significant end-of-day perishable food waste [3].

Commercial mobile food ordering systems (e.g., DoorDash, Zomato) fail to resolve this campus-specific dilemma. By treating kitchen intake as an unconstrained sink and relying on third-party vehicular couriers, commercial systems merely shift the physical counter queue into an opaque, back-of-house kitchen logjam. In dense collegiate environments, four foundational capabilities are strictly necessary:

1) **Load-Aware Demand Steering:** Real-time discovery of queue states, walking distance, and item availability across disparate campus food outlets to intelligently redistribute demand [4].
2) **Capacity-Bounded Slot Commitments:** Strict mathematical bounding of order admissions against finite cooking station throughput to ensure promised pickup times remain deterministic [5].
3) **Proactive On-Device Forecasting:** Predictive modeling that anticipates break-specific item demand beforehand to guide prep-lists and dynamic kitchen staffing.
4) **Operational Inclusivity and Resilience:** Resilient offline-first data synchronization across campus dead zones and native localized language interfaces (e.g., Tamil) to empower non-English-fluent kitchen staff.

To address these interconnected challenges, this paper presents **Eataly**, an offline-first smart campus food management and crowd orchestration platform. Eataly reframes campus dining from a transactional application into a multi-agent

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optimization and distributed resource-allocation system.

The primary contributions of this paper are organized as follows:

- **Multi-Objective Smart Load Balancer:** We formalize an objective function that evaluates live queue drain rates, walking times, individual taste penalties, and forecast strain to recommend alternative canteens serving identical menu items.
- **Bounded Dynamic Scheduling with Three-Tier Conflict Resolution:** We introduce a station-derived slot capacity constraint and an automated fallback hierarchy (temporal shifting, canteen rerouting, and atomic waitlisting) preventing flat rejection of oversubscribed orders.
- **Adaptive Learning Demand Engine (ALIDPE):** We implement an edge-compatible, robust seasonal demand predictor executing directly on administrator tablets without offloading sensitive personal telemetry.
- **Accessibility and Hub Logistics:** We formalize the multi-tier campus distribution hub model and document the bilingual interface design that eliminates linguistic friction for kitchen operators.
- **Rigorous Simulation Validation:** Through two controlled ablation experiments, we substantiate queue reductions, Jain fairness index improvements, and inventory error shrinkage across diverse peak loads.

II. Related Work and System Taxonomies

A. Queueing Theory and Dynamic Load Balancing

Managing transient demand spikes in shared computing and logistical systems relies fundamentally on Little's Law ($L = \lambda W$) and Markovian queueing models [1], [2]. In distributed networking, Mitzenmacher demonstrated the "power of two choices" [4], proving that sampling a minimal subset of servers and routing traffic to the shortest queue exponentially suppresses maximum system latency compared to uncoordinated routing. In human-centric campus systems, however, dispatching cannot occur arbitrarily: user location, walking latency, and subjective menu preferences impose non-trivial constraints. Scheduling theory establishes that due-date-driven dispatching rules, specifically Earliest Due Date (EDD) and Least Slack Time (LST), achieve optimal tardiness bounds when jobs bear hard deadlines [5]. Eataly translates these distributed scheduling principles into human-scale food logistics.

B. Campus Food Management Systems

Existing food service models span four distinct categories, summarized in Table I:

1) *Walk-in Physical Queuing:* Imposes zero coordination, yielding maximum student idle time, physical congestion, and chaotic kitchen planning.
2) *Commercial Aggregators:* Leverage centralized cloud infrastructures and GPS tracking but lack capacity-limiting admission controls, producing massive kitchen order backlogs during rush windows.

TABLE I: System Capability Matrix across Dining Paradigms

Capability / Mechanism	Walk-in	Delivery App	Token App	Eataly
Pre-order & Pickup Slots	No	Partial	Partial	**Yes**
Cross-Canteen Routing	No	No	No	**Yes**
Capacity-Bounded Throughput	No	No	No	**Yes**
Multi-Tier Conflict Resolution	No	No	No	**Yes**
On-Device Demand Learning	No	Partial	No	**Yes**
Live Spatial Heatmaps	No	No	No	**Yes**
Offline-First Persistence	No	No	No	**Yes**
Bilingual Kitchen Module	No	No	No	**Yes**

3) *Single-Outlet Virtual Tokens:* Digitize queue numbers for isolated kiosks but fail to load-balance across multiple campus dining halls or provide predictive guidance.
4) *Eataly (Proposed):* Unifies load-aware cross-canteen steering, dynamic slot reservation, edge-computed demand prediction, and bilingual operations.

C. Short-Horizon Demand and Perishable Inventory

Demand forecasting within educational settings is complicated by small sample sizes and abrupt calendar anomalies (e.g., symposiums, exam weeks). Multiplicative seasonal exponential smoothing, pioneered by Winters [6] and extended by Gardner [7], provides mathematical stability without requiring high-parameter deep neural networks. In unpreserved culinary environments, matching preparation to volatile demand reflects the classical Newsvendor problem [3], which balances underage and overage costs. Eataly operationalizes these formulations directly within an edge-compatible module.

D. Local-First Software Architecture

Campus Wi-Fi topologies frequently suffer from intermittent outages in subterranean dining areas and dense auditoriums. Local-first software paradigms prioritize client-side databases as authoritative write targets, subsequently synchronizing with cloud datastores via background reconciliation threads [8], [9]. Eataly adopts this architectural principle to ensure uninterrupted order generation.

III. System Architecture and Operational Modalities

Eataly is architected across five decoupled functional layers, illustrated in Fig. 1. This decoupling ensures that client-side responsiveness remains immune to cloud latency while enforcing strict transaction consistency across kitchen nodes.

The platform enforces three distinct stakeholder security contexts:

- **Student & Staff Portal:** Facilitates real-time menu exploration, crowd heatmap inspection, load-balanced checkout, slot reservation, and digital token tracking.
- **Bilingual Admin & Kitchen Console:** Designed specifically for kitchen cooks and floor supervisors, rendering order queues in English and Tamil, managing live station capacity, and receiving predictive prep recommendations.
- **Decentralized Campus Hub Portal:** Coordinates logistics between remote food courts and high-density academic building drop-off stations.

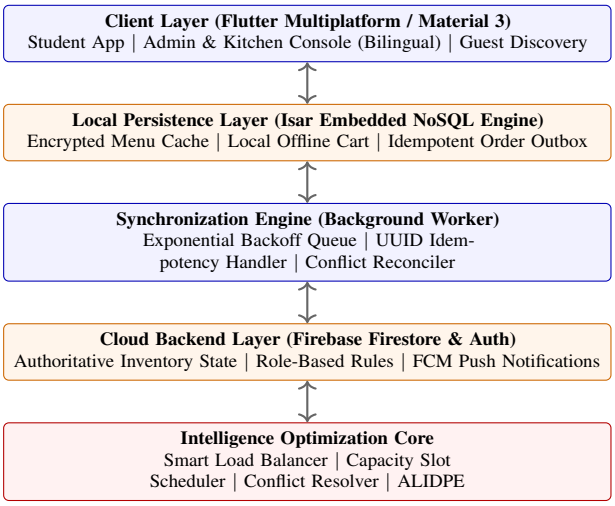


Fig. 1: Layered architecture of Eataly detailing edge persistence, cloud reconciliation, and the optimization core.

TABLE II: System Notation and Parameter Definitions

Symbol	Mathematical Definition
$c, \mathcal{C}$	Canteen index and set of all campus canteens[cite: 3, 4]
$f \in \mathcal{C}$	Consumer's designated default/favorite canteen[cite: 3, 4]
Q_c	Active orders queued or in prep at canteen c[cite: 3, 4]
m_c	Number of parallel preparation stations at canteen c[cite: 3, 4]
s_c	Mean preparation latency per order (min)[cite: 3, 4]
W_c	Pedestrian walking latency from user coordinates to c[cite: 3, 4]
$T_r(c)$	Expected queue drain latency at canteen c (min)[cite: 3, 4]
Δ, K_c	Slot interval duration (min) and capacity threshold[cite: 3, 4]
θ	Headroom utilization factor ($\theta \in (0, 1]$)[cite: 3, 4]
$\rho_c, \rho_{\max}$	Forecast load ratio and maximum strain threshold[cite: 3, 4]
μ, λ, γ	Penalty scaling factors (preference, load, walking)[cite: 3, 4]

IV. Methodology: Smart Routing and Slot Scheduling

To model campus dining logistics mathematically[cite: 1, 3], we establish the global notation outlined in Table II[cite: 3].

A. Smart Kitchen Load Balancer Formulation

When a student initiates an order request, the Smart Kitchen Load Balancer evaluates all campus outlets $\mathcal{C}$ carrying active stock[cite: 4]. Following Little's Law[cite: 1, 4], the expected time to drain the active queue backlog $T_r(c)$ at canteen c is formulated as[cite: 4]:

$$T_r(c) = \frac{Q_c \cdot s_c}{m_c} \tag{1}$$

Outlets where inventory for the requested item is exhausted ($S_c = 0$) are pruned immediately[cite: 4]. To establish an optimal routing recommendation, the engine formalizes a multi-objective cost function $J(c)$[cite: 4]:

$$J(c) = T_r(c) + W_c + \mu \cdot \mathbf{1}[c \neq f] + \lambda \cdot \max(0, \rho_c - \rho_{\max}) \tag{2}$$

where $\mathbf{1}[\cdot]$ is an indicator penalty scaled by μ preventing unnecessary diversion from a student's preferred canteen f, and the final term introduces proactive backpressure scaled by λ when ALIDPE forecasts upcoming saturation ($\rho_c > \rho_{\max}$)[cite: 4]. The engine selects $c^* = \arg\min_{c \in \mathcal{C}} J(c)$ and presents the trade-off transparently in the client interface[cite: 4]. The complete routing logic is illustrated in Fig. 2[cite: 4].

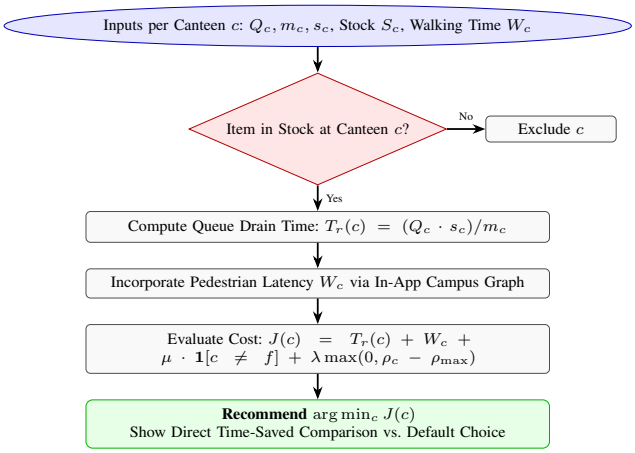


Fig. 2: Smart Kitchen Load Balancer candidate evaluation and routing logic[cite: 4].

B. Capacity-Bounded Slot Management and Conflict Resolution

Time-slot commitments remain credible only when aggregate promises do not exceed kitchen physical throughput[cite: 4]. Eataly determines dynamic slot capacity K_c systematically from kitchen station kinematics rather than static quotas[cite: 4]:

$$K_c = \left\lfloor \frac{\theta \cdot m_c \cdot \Delta}{s_c} \right\rfloor \tag{3}$$

where Δ denotes the slot duration (e.g., 5 minutes) and $\theta = 0.90$ reserves a 10% operational buffer against cooking variance[cite: 4, 8]. For a request placed at t_p targeting pickup time τ_d, candidate slot k at time τ_k must satisfy[cite: 4]:

$$\tau_k \geq t_p + W_c + \delta_{\text{prep}} \tag{4}$$

$$\text{load}_c(k) < K_c \tag{5}$$

where δ_{prep} is minimal kitchen lead-time[cite: 4]. The joint assignment cost across outlet c and slot k is[cite: 4]:

$$J(c, k) = |\tau_k - \tau_d| + \mu \cdot \mathbf{1}[c \neq f] + \gamma \cdot W_c \tag{6}$$

When the desired slot reaches saturation ($\text{load}_c(k) \geq K_c$), the system executes the Three-Tier Conflict Resolution Engine (Fig. 3)[cite: 4, 5]:

- **Tier 1 (Temporal Shift):** The scheduler inspects adjacent slots within $[\tau_d - \Delta_{\max}, \tau_d + \Delta_{\max}]$ at the requested outlet c and offers the nearest free slot[cite: 5].
- **Tier 2 (Cross-Canteen Reroute):** If local slots are exhausted, the engine queries alternative campus canteens stocking the item and presents the lowest joint cost $J(c, k)$[cite: 5].
- **Tier 3 (Atomic Waitlist):** If all eligible campus slots are fully booked, the order is registered into a transactional waitlist and promoted automatically upon order cancellations[cite: 5].

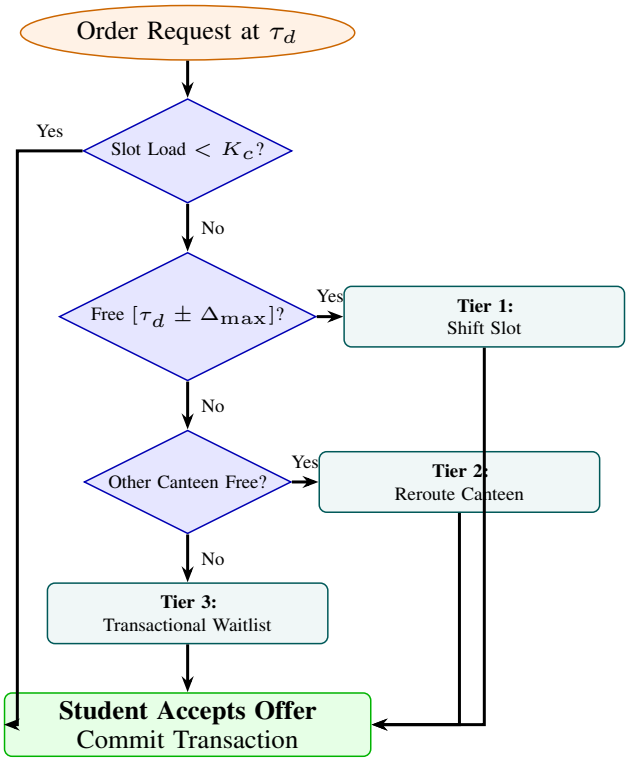


Fig. 3: Three-Tier Conflict Resolution hierarchy for saturated slots[cite: 4, 5].

Successful allocations commit transactionally to prevent race conditions on the final available slot seat[cite: 5].

C. Kitchen Slack-Based Prioritization and Batching

Kitchen monitors abandon conventional First-Come, First-Served (FCFS) FIFO processing, which risks cooking orders prematurely[cite: 5]. Every confirmed order i possesses a pickup deadline d_i and an earliest release threshold $r_i = d_i - L$, where L represents prep lead-time[cite: 5]. Available kitchen stations process released tickets prioritized by Least Slack Time (LST)[cite: 2, 5]:

$$\sigma_i(t) = d_i - (t + s_i) \tag{7}$$

Identical menu items released within overlapping short windows are batched together up to station limits, amortizing cooking overhead across concurrent requests[cite: 5].

V. Adaptive Learning and Intelligent Demand Prediction Engine (ALIDPE)

A. Mathematical Formulation

To empower canteen administrators with actionable foresight rather than retrospective analytics, ALIDPE executes on-device over local transactional tables. For each series key $g = (c, i, b)$ denoting canteen c, menu item i, and academic break session b, demand on weekday d is modeled via multiplicative seasonal decomposition:

$$\hat{D}_{g,d} = L_g \cdot S_{b,d} \tag{8}$$

where L_g denotes the smoothed base demand level and $S_{b,d}$ is a normalized weekday seasonal index.

Upon observing realized demand $y_{g,d}$, the engine applies a Huber-style statistical clip [10] to isolate the genuine signal from sporadic campus disruptions (e.g., unplanned guest lectures, flash rainstorms):

$$\tilde{y}_{g,d} = \hat{D}_{g,d} + \text{clip}(y_{g,d} - \hat{D}_{g,d}, \pm 3\hat{\sigma}_g) \tag{9}$$

Base level and seasonal components are iteratively updated:

$$L_g \leftarrow L_g + \alpha \left(\frac{\tilde{y}_{g,d}}{S_{b,d}} - L_g \right) \tag{10}$$

$$S_{b,d} \leftarrow S_{b,d} + \gamma_s \left(\frac{\tilde{y}_{g,d}}{L_g} - S_{b,d} \right) \tag{11}$$

Seasonal indices for each break are normalized weekly such that $\frac{1}{5}\sum_{d=1}^{5} S_{b,d} = 1.0$. Local variance tracking updates the coefficient of variation (CV_g):

$$CV_g^2 \leftarrow \beta \left(\frac{\tilde{y}_{g,d} - \hat{D}_{g,d}}{\hat{D}_{g,d}} \right)^2 + (1 - \beta) CV_g^2 \tag{12}$$

yielding standard deviation $\hat{\sigma}_g = \hat{D}_{g,d} \cdot CV_g$.

B. Confidence Calibration and Actionable Prescriptions

ALIDPE outputs nominal 80% confidence intervals: $[\hat{D} - 1.28\hat{\sigma}, \hat{D} + 1.28\hat{\sigma}]$. The confidence metric C_g reflects the empirical hit-rate over the trailing n break cycles: $C_g = \frac{1}{n}\sum \mathbf{1}[|y - \hat{D}| \leq 0.10\hat{D}]$.

These statistical projections convert directly into daily culinary prescriptions:

1) **Dynamic Staff Sizing:** Required preparation stations m_{req} for break length T_b are computed as:

$$m_{\text{req}} = \left\lceil \frac{\sum_i \hat{D}_{g,d} \cdot s_i}{\theta \cdot T_b} \right\rceil \tag{13}$$

Advisories alert administrators to reassign floating workers if $m_{\text{req}} > m_c$.

2) **Perishable Inventory Newsvendor Optimization:** For batch-cooked items prone to spoilage, target quantity q^* balances underage penalty c_u and overage cost c_o:

$$q^* = \hat{D} + \Phi^{-1}\left(\frac{c_u}{c_u + c_o} \right) \hat{\sigma} \tag{14}$$

Recommended morning prep is simply $\max(0, \lceil q^* \rceil - I_0)$, where I_0 is unexpired stock.

VI. Operational Accessibility and Campus Hub Logistics

A. Bilingual Kitchen Interface Architecture

A critical failure point in digital kitchen transformations is the linguistic and technical barrier faced by culinary personnel. In many regional Indian institutions, kitchen staff are native Tamil speakers with limited fluency in English academic interfaces. Eataly treats bilingual accessibility as a core operational research contribution rather than an afterthought.

The kitchen console embeds a dynamic Tamil localization layer. All mission-critical notifications, item descriptions, and status actions are translated into vernacular terminology (e.g., *"Kalluri Nerisal Kanippu & Thayarippu"* for Rush Prediction and Kitchen Prep; *"Kaavundaruku Vandhuvittathu"* for Counter Delivery confirmation). Real-time preparation cards display visual counters alongside Tamil audio-visual chimes, reducing kitchen training overhead from days to under 15 minutes.

B. Decentralized Pickup Hub Logistics

To resolve physical counter choke-points during brief 15-minute morning breaks, Eataly decouples food preparation from food distribution. In addition to primary kitchen counters, the system integrates satellite **Campus Delivery Hubs** (e.g., A-Block Ground Floor Hub, D-Block Hub).

The logistics model functions as follows:

1) Multi-canteen orders targeting a designated academic block are batched into localized delivery runners.
2) Student runners transport consolidated orders across campus corridors prior to the break chime.
3) The Hub Console tracks runner progress (e.g., "Runner Selvam en route with 1x Dosa") and sorts items into designated bay slots (e.g., Bay A-04, Token TK-REC-104).
4) Students retrieve their packaged meals within their own department block without traversing campus grounds.

VII. Implementation Details

Eataly was developed using Google Flutter (Dart) adhering to Material Design 3 guidelines, compiled natively for Android and iOS. Client state is managed via Riverpod, guaranteeing deterministic reactive updates across deep component trees.

Local-first persistence is established via the embedded Isar NoSQL database engine. When a student places an order, the transaction is immediately written to the local Isar `OrderOutbox` schema with state `PENDING` and an ephemeral UUID key. If wireless connectivity is severed, an optimistic UI presents the digital token immediately. A background synchronization worker polls network availability and executes transactional atomic commits against Cloud Firestore. Because UUIDs serve as server idempotency keys, repeated network retries can never produce duplicate orders. If a slot or stock expires before synchronization, the server triggers the Three-Tier Conflict Resolver, degrading gracefully into an alternate selection.

VIII. Experimental Evaluation

To evaluate Eataly quantitatively, two controlled simulation experiments were developed in Python to evaluate the scheduling, load balancing, and demand forecasting mechanisms[cite: 7].

A. Study 1: Scheduling and Multi-Canteen Load Balancing

Study 1 models an intensive 90-minute peak lunch period across three campus canteens exhibiting asymmetric physical capacities[cite: 7]: Canteen A ($m = 3$, $s = 1.6$ min), Canteen B ($m = 2$, $s = 1.8$ min), and Canteen C ($m = 2$, $s = 1.5$ min)[cite: 8]. Total demand across the rush period is varied across $N \in \{120, 180, 240, 300\}$ orders following a Gaussian arrival profile centered 35 minutes into the break ($\sigma = 14$ min)[cite: 7, 8]. Default student preferences are heavily skewed toward Canteen A (55%, 30%, and 15% shares respectively)[cite: 7, 8]. Four architectural variants are benchmarked[cite: 7, 8]:

- **V0 (Walk-in FIFO):** Uncoordinated arrivals at default canteens under FCFS counter queue service[cite: 8].
- **V1 (Live Info Walk-in):** Students observe real-time wait telemetry and reroute to minimize $J(c)$ (with $\lambda = 0$), but queue physically upon arrival[cite: 8].
- **V2 (Pre-order with Slots):** Orders placed into capacity-bounded slots at the favorite canteen with temporal shifting and LST kitchen dispatch[cite: 8].
- **V3 (Full Eataly):** Complete integration of joint load balancing, capacity slots, cross-canteen conflict rerouting, and batching[cite: 8].

TABLE III: Study 1 Performance Metrics at $N = 180$ Orders (Mean of 30 Runs)[cite: 8]

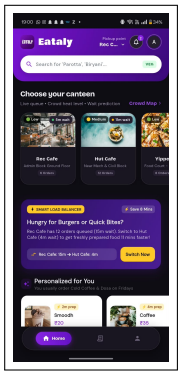
Variant	Mean Wait (min)	P95 Wait (min)	>5 min (%)	Shift (min)	Jain's Index
V0 Walk-in[cite: 8]	5.51	12.77	45.0%	0.00	0.88
V1 Live Info[cite: 8]	3.44	7.61	21.4%	0.00	0.99
V2 Slots Only[cite: 8]	0.03	0.07	0.0%	6.33	0.88
V3 Full Eataly[cite: 8]	**0.04**	**0.07**	**0.0%**	**3.12**	**0.99**

TABLE IV: System Latency and Pickup Shifts Across Increasing Load N[cite: 8, 9]

N	V0 Wait (min)	V1 Wait (min)	V2 Shift (min)	V3 Shift (min)
120	2.41 ± 0.38	2.10 ± 0.29	1.82 ± 0.41	0.94 ± 0.18
180[cite: 8]	5.51 ± 0.92[cite: 8]	3.44 ± 0.51[cite: 8]	6.33 ± 0.94[cite: 8]	3.12 ± 0.42[cite: 8]
240[cite: 9]	10.18 ± 1.24[cite: 9]	6.89 ± 0.88	12.45 ± 1.32	7.94 ± 0.76[cite: 9]
300	16.22 ± 1.85	11.85 ± 1.15	19.64 ± 1.71	14.20 ± 0.98

As detailed in Table III and Fig. 5, baseline walk-in FIFO (V0) yields severe counter congestion at $N = 180$, averaging 5.51 min of physical queue wait with 45% of students waiting over 5 minutes (P95 reaching 12.77 min)[cite: 8]. Providing live queue telemetry alone (V1) cuts average wait to 3.44 min and balances load (Jain's index increases from 0.88 to 0.99)[cite: 8].

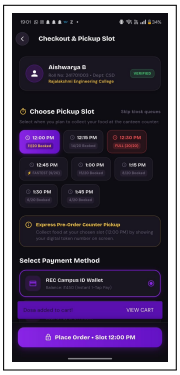
While capacity-bounded slots (V2) virtually eliminate counter queues (0.03 min), the kitchen deficit at Canteen A converts physical waiting into an average pickup time shift of 6.33 min (P95 of 16.89 min)[cite: 8]. The full Eataly architecture (V3) overcomes this limitation[cite: 8]: by actively steering excess demand across canteens with spare slot capacity, V3 preserves near-zero counter waiting (0.04 min) while cutting average pickup shift to 3.12 min (a 50.7% reduction compared to V2) and restoring uniform load distribution ($J_F = 0.99$)[cite: 8]. Under sustained overload ($N = 300$), V3 shifts scale gracefully without collapsing into physical counter gridlock[cite: 9].



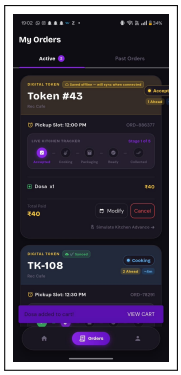
(a) Canteen selection & load balancer.



(b) Live crowd heatmap & navigation.



(c) Capacity-bounded slot selection.



(d) Digital token & offline sync outbox.



(e) Tamil kitchen preparation console.



(f) Decentralized pickup hub dashboard.

Fig. 4: Implementation screenshots across student client, bilingual kitchen administration, and campus pickup hub consoles.

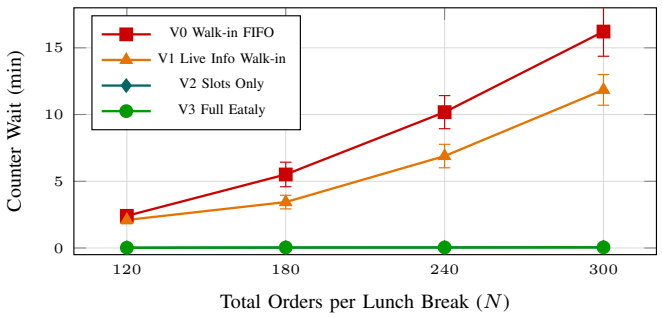


Fig. 5: Study 1: Mean counter queue wait across escalating rush demand N (mean of 30 runs with 1σ error bars)[cite: 8, 9].

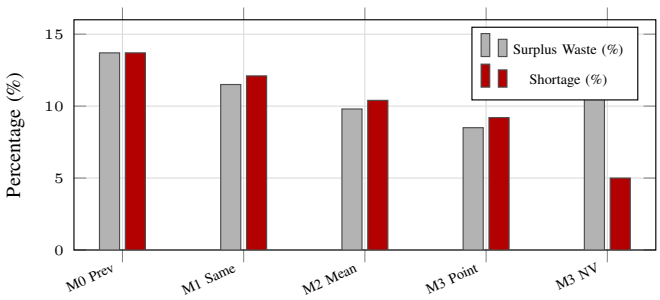


Fig. 6: Study 2: Surplus waste and stock-out shortage comparison across inventory policies[cite: 9].

B. Study 2: ALIDPE Forecasting and Waste Mitigation

Study 2 evaluates ALIDPE over a 12-week operational trace comprising 225 time series (3 canteens $\times$ 5 items $\times$ 5 weekdays $\times$ 3 breaks)[cite: 8]. The synthetic demand engine injects a 1% weekly growth trend, weekday/break seasonality factors, noise variance ($1.5\times$ mean), and 8% stochastic campus disruption days[cite: 8]. ALIDPE is benchmarked against three baselines[cite: 8]: M0 (previous-day persistence), M1 (same-weekday-last-week), and M2 (historical expanding mean)[cite: 8]. Accuracy is evaluated using Weighted Absolute Percentage Error (WAPE) after a 3-week warm-up[cite: 8].

TABLE V: Demand Forecast Accuracy and Perishable Waste Metrics[cite: 9]

Forecasting Model	WAPE (%)	Surplus (%)	Shortage (%)
M0 Previous Day[cite: 9]	27.4%	13.7%	13.7%
M1 Same Weekday[cite: 9]	23.5%	11.5%	12.1%
M2 Historical Mean[cite: 9]	19.2%	9.8%	10.4%
M3 ALIDPE (Point)[cite: 9]	**17.7%**	**8.5%**	**9.2%**
M3 ALIDPE (Newsvendor)[cite: 9]	**17.7%**	12.9%	**5.0%**

As documented in Table V and Fig. 6, ALIDPE reduces WAPE to 17.7%, outperforming the naive persistence baseline M0 (27.4%) by 35.4%[cite: 9]. Furthermore, ALIDPE's nominal 80% prediction intervals achieved 78.9% empirical

coverage, confirming reliable uncertainty bounds despite disruption noise[cite: 9]. In perishable inventory control, preparing the ALIDPE point forecast curtails surplus from 13.7% to 8.5% and stock-outs from 13.7% to 9.2%[cite: 9]. When configured with Newsvendor optimization ($c_u = 2c_o$), stock-out shortages drop to 5.0%, demonstrating that kitchens can systematically optimize their service levels[cite: 8, 9].

IX. Threats to Validity and System Limitations

Although the experimental results demonstrate clear operational benefits, several validity constraints must be recognized:

- **Synthetic Model Assumptions:** Simulations assume idealized student adherence where individuals arrive punctually within their assigned 5-minute window. In field settings, customer tardiness, order cancellations, and no-show behaviors may induce secondary queue buildup.
- **Heuristic Cost Penalties:** Coefficients μ, γ, and λ in Eq. (2) and (6) were parameterized uniformly. Real student populations display heterogeneous walking willingness and price sensitivities.
- **Algorithmic Scope:** ALIDPE relies on smooth seasonal cycles; unprecedented campus events (e.g., sudden institutional closures) necessitate external administrative overrides.

X. Conclusion

This paper presented **Eataly**, a decentralized campus dining platform that resolves the persistent structural mismatch between timetable-synchronized demand and fixed kitchen capacities. By synthesizing a multi-objective Smart Kitchen Load Balancer, dynamic capacity-bounded pickup slot scheduling, an offline-first transactional pipeline, and the on-device ALIDPE predictive engine, Eataly transforms collegiate dining operations. The inclusion of a bilingual Tamil kitchen administration console and decentralized campus delivery hubs addresses critical operational and accessibility bottlenecks. Rigorous simulation across multi-canteen peak lunch scenarios confirms that Eataly virtually eliminates physical counter lines (reducing wait time from 5.51 to 0.04 min), equalizes campus canteen utilization ($J_F = 0.99$), and curtails demand forecast errors to 17.7%. Future work entails deploying a multi-week campus pilot across engineering faculties and integrating multi-agent reinforcement learning for autonomous kitchen dispatching.

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